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OLED: Yet another technology!

Canon and Toshiba are jointly developing yet another related technology called organic light-emitting diode (OLED). OLED is a film-based carbon technique that has so far been used only in handheld prototypes (besides some smaller-size TV prototypes). But OLED screens are expensive, which leaves NED as potentially the cheapest and best alternative to HDTV, CRT and LCD screens.

NEDs: Why are they the best alternative?

A nano-emissive display is a high-resolution, full-colour display. It is also light-weight and physically thin. The display uses carbon nanotube technology. Carbon nanotubes are grown directly on the glass of the display system. This makes the display very energy-efficient in the way it emits electrons (which excite the coloured phosphors to emit visible light). Instead of using a CRT or millions of tiny LED lights to project a video image, NED uses millions of accelerated electrons charged by just 5 to 10 volts of electricity, compared to 5000 volts for large-screen, high-definition LCDs.

The electrons shoot toward a phosphor plate, creating the moving image.

The first working prototype of the NED by Motorola is approximately 3mm (1/8 inch) thick and measures about 12.7 cm (5 inches) diagonally. Engineers anticipate that multiple sections of small size will be combined to produce large displays for computers, high-definition television (HDTV) receivers and even road signs. This technique requires lesser voltage than a CRT, so the displays won't consume as much power. And, unlike LCDs, a nano-emissive display, which uses carbon nanotube technology, will be easily viewable from all angles.

So advantages of NEDs include relatively low cost of manufacture, less power consumption, low-voltage operation, vivid colours, wide viewing angle and high image resolution. Another advantage is that there is, at least in theory, no limit to how large a display can be made using many small sections. Some say that these televisions could last more than twelve years under normal use, which is a fairly good lifespan for televisions with newer displays.

Other research efforts aimed at using carbon nanotubes to fabricate displays use a less efficient pasting method to attach the nanotubes to the glass. It is believed that this solution is

less expensive than field-emissive and plasma displays.

Future developments

Nano-emissive displays are said to have better quality, durability and economics than most traditional forms of display technology. Combining these three factors into one product is considered the triple crown of display technology and may offer the company that is able to mass produce the product a significant market advantage over other technologies. Further, if other companies also choose NED, they could start manufacturing high-definition sets very early and at highly competitive prices.

However, there are several problems that must be solved before NED displays become widely available. Using carbon nanotubes has several challenges. First, these need to be grown in a uniform and consistent fashion. Second, the glass display needs to be sealed properly in order to prevent impurities from corrupting the image quality. Third, phosphor coatings, which provide the colours, must be functional and reliable over long periods of time. Manufacturers would need to maintain a high level of quality in order to keep production costs low.

Once nano-emissive displays reach the mass market, these are expected to provide superior quality in a number of different ways to display technologies currently available. ●

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